COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
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ACTIVITY NO. _6.1_ Linear and Binary Search

1. PROCEDURE OUTPUT

What procedures did you perform?

1. What is a search tree in data structures?

- What makes a search tree special is because it is a node-based tree data structure that is designed for efficiently locating, deleting and inserting specific keys from a set.

2. What are the Different types of search algorithm in data structures? Differentiate each type of search.

- **Linear Search:**
  - It functions in one by one comparison of the item to be searched with array elements, it is only used in an unsorted array. Additionally, it takes linear or O(n) time.

- **Binary Search:**
  - It functions by comparing the array's middle element first and if the middle element is the same as the input, then it will return. Otherwise it searches in either the left half or right half based on the comparison result. This is used for a sorted array and it takes O (Log n).

3. What operations / implementations can be performed using binary and linear search operations?

- Linear search operations can be implemented in multidimensional array while Binary Search can be implemented in only one single dimensional array.

4. What are the advantages in using binary search tree as data structure?

- The advantage of using a binary search tree as a data structure is that it is a very fast process

5. Give an example program using binary search and Linear search.

- An example program using binary search is an operating system file finder, it uses a search bar in windows explorer or mac finder that uses binary search variants to immediately show what you're searching for as you type. On the other hand, an example program using linear search is a music playlist shuffling, where a shuffled playlist on Spotify hits the "next" button, it uses a linear search to check what is the next randomly shuffled song that is in the queue.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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